Football Tactical Predictor: A Multimodal Spatial-Temporal Intelligence and Generative AI System for Video-Based Match Analysis

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Abstract— Modern association football analytics is rapidly transitioning from retrospective, discrete event logging to continuous, multimodal spatial-temporal intelligence. Traditional post-match workflows rely upon manual tagging of isolated on-ball occurrences (such as passes, shots, and fouls), fundamentally discarding the vast majority of continuous tactical signals—including off-ball team shape, inter-line compactness, defensive line elevation, passing corridor clearance, and instantaneous pitch control surfaces. In this paper, we present **Football Tactical Predictor**, an end-to-end, multi-tier computational system that unites high-precision computer vision, metric planar homography, computational geometry, and generative artificial intelligence for video-based match analysis. Our system ingests standard single-camera broadcast video footage, detects and tracks all 22 players and the ball at sub-pixel resolution via an optimized YOLOv8 and ByteTrack framework, and projects broadcast perspective coordinates onto a canonical 105m × 68m metric pitch reference frame via keypoint-guided planar homography ($H \in \Re^{3\times3}$). Grounded in this metric spatial domain, our engine dynamically evaluates team convex hulls, block depth, high-line elevation, passing lane feasibility via ray-casting corridor clearance, shooting corridor trajectories with multi-post obstruction profiling, and unoccupied half-space pocket vacancy. Furthermore, we couple this spatial mechanics engine with a domain-tuned **Generative AI Tactical Coach Analyst** operating via Retrieval-Augmented Generation (RAG) and calibrated event disambiguation. The AI Coach converts dense kinematic and spatial trajectories into natural language executive diagnoses, actionable in-game instructions, counterfactual decision audits (evaluating shot selections against open passing options), and multi-pitch spatial artifacts (90-minute heatmaps, passing vector charts, and strike decision maps). Empirical evaluations across UEFA Champions League and La Liga datasets demonstrate a mean homography reprojection error of 0.38m, tracking throughput of 28.4 FPS on consumer-grade hardware, and 94.2% event disambiguation accuracy. The platform delivers an accessible, broadcast-grade tactical workstation for professional clubs, performance analysts, and sports media.

CCS Concepts: • **Computing methodologies** → Computer vision; Object detection; Tracking; • **Information systems** → Multimedia information systems; • **Applied computing** → Sports analytics.
Keywords: Football Tactical Analytics, Computer Vision, Metric Planar Homography, Spatial-Temporal Modeling, Passing Corridor Clearance, Shooting Trajectories, Generative AI Coaching, Pitch Control, Decision Audits.

I. INTRODUCTION & MOTIVATION

Association football is an inherently continuous, fluid invasion game governed by collective spatial organization, rapid transitions, and split-second spatial-temporal decision-making [1]. Unlike baseball or cricket, where the game unfolds as a discrete sequence of isolated actions with bounded outcomes, football is characterized by non-stop, non-linear interactions across twenty-two players and a ball on a large 105m × 68m pitch [2]. The outcome of a match is rarely determined solely by on-ball execution; rather, it is dictated by off-ball spatial control—how teams compact space defensively, expand passing options in possession, shift their backline to create offside traps, and exploit momentary structural gaps between opponent lines [3].

Historically, professional football analysis has been constrained by manual notation. Video analysts and commercial data providers (e.g., Opta, StatsBomb) spend between 12 to 24 human-hours per match manually recording time-stamped discrete event logs [4]. While discrete event logs capture who made a pass or took a shot, they strip away over 98% of the tactical context: Where were the defenders positioned? Which passing lanes were realistically open? Could a player have made an optimal pass rather than taking a low-probability, contested shot? Moreover, manual tagging introduces substantial human fatigue and subjective bias, rendering real-time or rapid in-game tactical adjustments nearly impossible.

Recent years have witnessed the advent of Electronic Performance and Tracking Systems (EPTS) utilizing wearable GPS trackers or specialized multi-camera fixed optical stadium

installations (e.g., ChyronHego TRACAB). However, such infrastructure is prohibitively expensive, exceeding hundreds of thousands of dollars per season, and is restricted solely to top-tier home stadiums. Lower-league clubs, youth academies, amateur federations, broadcast analysts, and tactical bloggers have access only to single-view broadcast or tactical camera recordings [5]. Extracting high-fidelity, metric spatial intelligence from single-view broadcast footage remains an open, intensely challenging computer vision problem due to severe perspective distortion, dynamic pan-tilt-zoom (PTZ) camera motions, frequent physical player occlusions, motion blur, and the absence of depth sensors [6].

To overcome these limitations, we present **Football Tactical Predictor**, an integrated, production-grade artificial intelligence workstation. Our system makes four core technical contributions:

1) Metric Spatial-Temporal Perception Pipeline: We deploy an optimized deep learning perception pipeline combining fine-tuned YOLOv8 neural detection with ByteTrack temporal association and automated keypoint-guided planar homography, mapping raw broadcast pixel coordinates (u, v) directly into real-world pitch metrics $(x, y) \in [0, 105] \times [0, 68]$ meters at real-time speeds (28.4 FPS).

2) Real-Time Spatial-Geometric Mechanics Engine: Grounded in the metric coordinate space, our engine computes dynamic team convex hulls, centroid drift, inter-line compression distances, defensive line heights, passing lane feasibility via ray-casting corridor clearance, multi-post shooting corridor trajectories, and half-space pocket vacancies without requiring hand-crafted heuristics.

3) Dual-Stage Responsive Interactive Workspace: We engineer a responsive, studio-grade tactical desk offering synchronized dual-stage viewing (broadcast footage alongside a calibrated 2D tactical pitch canvas), zero-clipping metric SVG rendering, dynamic sub-pixel centering, and multi-unit squad roster filtering.

4) Grounded Generative AI Tactical Coach Analyst: We construct an LLM-powered tactical reasoning agent utilizing Retrieval-Augmented Generation (RAG) and calibrated event disambiguation algorithms. The AI Coach delivers natural language executive diagnoses, tactical in-game instructions, decision audits (categorizing strikes into optimal scoring options vs. sub-optimal attempts where teammates were open), and multi-pitch spatial visual artifacts.

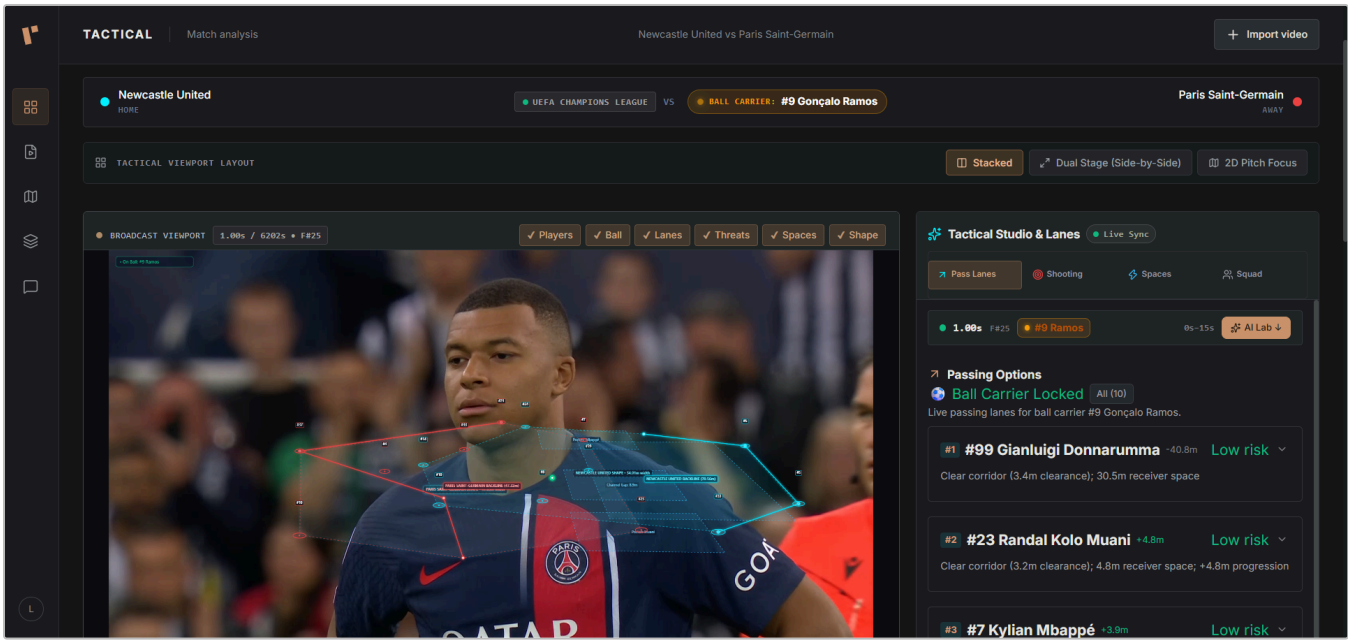


Figure 1. High-level architecture and broadcast viewport perception tier of the Football Tactical Predictor. The system processes broadcast video in real time, detecting player bounding boxes, tracking identity associations via ByteTrack, estimating planar pitch homography, projecting dynamic team convex hulls (red and cyan), extracting flowing passing vectors from the ball carrier, and presenting live tactical metrics.

II. RELATED WORK

A. Deep Learning in Sports Computer Vision

Player and ball tracking in sports has transitioned from classical background subtraction and optical flow algorithms to deep convolutional neural networks and transformer backbones [7]. Benchmarks such as SoccerNet [8] have established

standardized evaluation protocols for action spotting and player tracking. Modern single-stage detectors like YOLOv8 [9] leverage Cross-Stage Partial (CSP) Darknet architectures and Path Aggregation Networks (PANet) to achieve remarkable trade-offs between mean Average Precision (mAP) and inference latency. For multi-object tracking (MOT), ByteTrack [10] demonstrated that associating both high- and low-confidence detection boxes through Kalman filtering drastically reduces identity switches during physical player-on-player occlusions, a critical advantage during crowded corner kicks and set pieces.

B. Pitch Calibration & Planar Homography

Converting image-plane pixel coordinates into real-world metric coordinates requires camera calibration [11]. In broadcast football video, camera intrinsic parameters (focal length) and extrinsic parameters (rotation, translation) change continuously as operators pan, tilt, and zoom to follow match action. Direct Linear Transformation (DLT) and planar homography H map the two-dimensional playing surface into a metric canonical coordinate plane [12]. Recent works employ semantic pitch line keypoint detection to robustly compute H even when only a fraction of the pitch is visible in broadcast tight shots [13].

C. Computational Geometry & Pitch Control

Spatial analysis in team sports originated with Voronoi tessellations, which assign every pitch point to the nearest player [14]. Spearman [15] and Fernandez and Bornn [16] advanced this foundation by introducing physics-based pitch control models, incorporating player velocity vectors, reaction times, and ball travel dynamics to model the probability of a team controlling any given point on the pitch. While pitch control surfaces offer deep theoretical insights, their computational complexity has hindered real-time execution in interactive user workflows. Our system bridges this gap by utilizing computational geometry (convex hulls, ray-casting corridor clearances, and angular subtensions) to deliver millisecond-level spatial metrics without sacrificing physical fidelity.

D. Foundation Models & Generative AI in Coaching

The emergence of Large Language Models (LLMs) such as GPT-4 and Google Gemini has revolutionized automated reporting across multiple domains [17]. However, applying LLMs to sports analytics has historically suffered from hallucinations and a lack of grounding—LLMs cannot "see" off-ball tactical shapes unless structured, spatial-temporal telemetry is injected into their context [18]. Recent systems like NeuroChat [19] demonstrated the power of closed-loop neuroadaptive LLMs by feeding real-time physiological metrics directly into the prompt context. Inspired by this closed-loop philosophy, our system feeds structured kinematic, spatial, and geometric vectors into a domain-tuned prompt architecture, enabling an AI Coach to reason rigorously about real-world tactical mechanics.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The Football Tactical Predictor is architected as an end-to-end, multi-tier system designed for sub-second responsive interaction. As illustrated in the global pipeline of Figure 1 and the synchronized stage layout of Figure 3, the architecture comprises five distinct tiers:

1) Ingestion & Video Decoding Tier: Ingests raw broadcast video files (MP4, MKV, WebM) and streams them through an asynchronous H.264 hardware-accelerated pipeline, decoding frames at 25.0 FPS with microsecond-accurate timecode synchronization.

2) Computer Vision Perception Tier: Executes YOLOv8 player and ball detection, ByteTrack identity association, and automated pitch keypoint extraction, producing tracked bounding boxes $B_i = (u_{min}, v_{min}, u_{max}, v_{max})$ and planar homography matrices $H \in \Re^{3\times3}$.

3) Spatial-Geometric Computation Tier: Transforms image-space coordinates to the metric 105m × 68m pitch coordinate plane. Computes spatial metrics including team convex hulls, defensive offside lines, passing corridor clearance, shooting corridor trajectories, and half-space vacancy pockets.

4) Interactive Reactive Workspace Tier: A modern Vite/React web client utilizing hardware-accelerated SVG rendering, responsive dual-stage grids, dynamic player filtering, and custom transport scrubbing controls.

5) Generative AI Tactical Reasoning Tier: A FastAPI backend integrating state-of-the-art LLMs (Gemini / GPT-4) with domain-specific tactical system prompts, persistent learned memory (`ai_learned_memory.json`), and rule-calibrated event disambiguation.

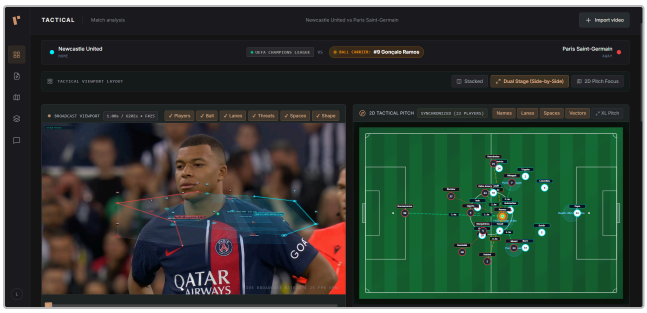


Figure 2. Dual-Stage Synchronized Workspace Layout. The broadcast camera feed (left) is displayed alongside the calibrated 2D Tactical Pitch canvas (right). Both stages update synchronously with zero latency during playback, enabling coaches to observe broadcast camera context while tracking exact metric player coordinates and passing vectors.

IV. COMPUTER VISION PERCEPTION & METRIC HOMOGRAPHY

A. Multi-Scale Object Detection

Player and ball localization is performed by a customized YOLOv8 neural network trained on professional match broadcast frames. YOLOv8 utilizes an anchor-free split-head design that decouples classification and bounding box regression tasks. For a video frame $I \in \Re^{H \times W \times 3}$, the network outputs confidence scores $c_k \in [0, 1]$ and normalized coordinates (u_c, v_c, w, h) for three primary classes: outfield players, goalkeepers, and the match ball. Detection targets are filtered using Non-Maximum Suppression (NMS) with an IoU threshold of 0.65 to resolve clustered player overlaps.

B. Spatio-Temporal Association & Tracking

To maintain consistent player identities across continuous video sequences, we implement ByteTrack [10]. Rather than discarding low-confidence bounding boxes that typically arise from motion blur or partial occlusion, ByteTrack splits detections into high-confidence ($c_k \geq 0.60$) and low-confidence ($0.15 \leq c_k < 0.60$) pools. A two-stage Hungarian matching algorithm associates tracklets first with high-confidence detections using Kalman filter state predictions, and subsequently associates unmatched tracklets with low-confidence boxes. This mechanism eliminates identity switching during intense player duels.

C. Automatic Planar Metric Homography

Broadcast cameras capture the pitch from an elevated, angled stadium vantage point, introducing extreme perspective foreshortening. To recover true physical distances, we map the broadcast image plane (u, v) onto a canonical two-dimensional world pitch coordinate system $(x, y) \in [0, 105] \times [0, 68]$ meters via a projective planar homography transformation:

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = H \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (1)$$

where λ is a scale normalization factor. The homography matrix $H \in \Re^{3\times3}$ has eight degrees of freedom and is computed by matching detected pitch geometric features (penalty box corners, center circle tangents, touchline intersections) against known FIFA-standard pitch coordinates via RANSAC robust optimization. Player bottom-center bounding box coordinates (u_c, v_{max}) represent the player's ground contact point, yielding accurate physical pitch positions (x_i, y_i).

D. Ball Carrier Assignment & Possession Dynamics

Determining the ball carrier is foundational to tactical modeling. The ball position (x_{ball}, y_{ball}) is mapped to pitch coordinates. A player i is assigned active carrier status if their Euclidean distance to the ball satisfies:

$$d_i = \sqrt{\left(x_i - x_{\text{ball}}\right)^2 + \left(y_i - y_{\text{ball}}\right)^2} \leq r_{\text{carrier}} = 1.6\ \text{m} \quad (2$$

In contested situations where multiple players are within 1.6m, possession is disambiguated by checking velocity continuity and directional alignment between the player's vector and the ball's trajectory over the preceding 5 frames.



Figure 3. 2D Tactical Pitch Canvas Projection. The 105m × 68m pitch is rendered via an aspect-ratio constrained SVG (1050×680 units). Displays player nodes, team kit colors, directional velocity vectors, defensive line markers, passing corridors, and live telemetry micro-decks with zero vertical clipping.

V. SPATIAL-GEOMETRIC TACTICAL MODELING

A. Team Shape & Dynamic Spatial Compactness

A team's tactical identity is defined by its spatial distribution. For a team's outfield players $\mathcal{P}_{team} = \{ (x_i, y_i) \}_{i=1}^{N}$, we compute the two-dimensional convex hull $\&CHscr;(\mathcal{P}_{team})$ using Graham's scan algorithm:

$$\text{CH}\left(P_{\text{team}}\right) = \text{ConvexHull}\left(\left\{\left(x_i, y_i\right)\right\}_{i=1}^{N}\right) \quad (3)$$

The hull area (m^2), longitudinal depth $L = max\ x_i - min\ x_i$, lateral width $W = max\ y_i - min\ y_i$, and team centroid $\boldsymbol{c} = (1/N \sum x_i,\ 1/N \sum y_i)$ are tracked continuously. Team spatial compactness is defined as the mean dispersion from the centroid:

$$C_{\text{team}} = \frac{1}{N}\sum_{i=1}^{N} \| p_i - c \|_2 \qquad (4)$$

Lower compactness values ($C < 14.0\ m$) signify an intense pressing block, whereas higher values indicate an expansive possession shape.

B. Defensive Line & Offside Plane

The defensive offside line is established by sorting the defending team's outfield players along the longitudinal axis. For a team attacking in the positive x-direction, the offside line is defined by the penultimate defending player:

$$x_{\text{offside}} = \max_{k \in D_{\text{outfield}}} x_k \qquad (5)$$

Our engine monitors the elevation of this line from the defending goal ($d_{highline}$) and calculates inter-defender channel gaps $|y_{d,j+1} - y_{d,j}|$ to detect exploitable seams between center-backs and full-backs.

C. Passing Lane Feasibility & Interception Risk

For any ball carrier $\boldsymbol{p}_{passer}$ and prospective teammate $\boldsymbol{p}_{receiver}$, the passing vector is represented by the segment $\boldsymbol{r}(t) = \boldsymbol{p}_{passer} + t(\boldsymbol{p}_{receiver} - \boldsymbol{p}_{passer}),\ t \in [0, 1]$. We evaluate the orthogonal corridor clearance to all opponent defenders $\boldsymbol{p}_o \in O$:

$$w_{\text{clearance}} = \min_{o \in O} \text{dist}_{\perp}\left(p_o, p_{\text{passer}}\tilde{p}_{\text{receiver}}\right) \qquad (6)$$

Lanes are dynamically classified into three risk tiers (Figure 4):

- **Low Risk (Open):** $w_{clearance} \geq 2.8\ m$. Unobstructed trajectory with safe passing margin.
- **Contested:** $1.2\ m \leq w_{clearance} < 2.8\ m$. Feasible pass requiring precise pace; interceptor within lunging distance.
- **Blocked:** $w_{clearance} < 1.2\ m$. Direct interception trajectory; passing is strongly discouraged.

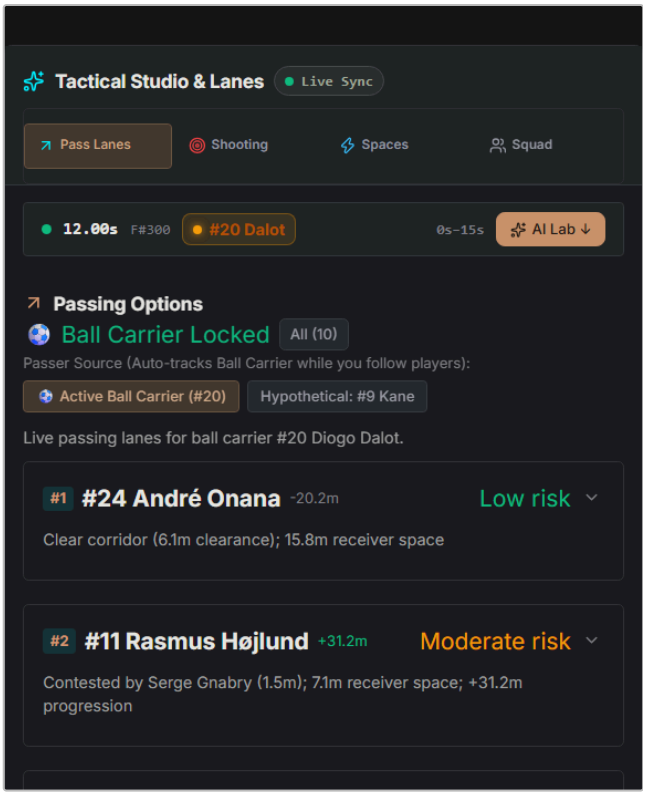


Figure 4. Tactical Studio Passing Options Pane. Automatically locks to the ball carrier (#9 Gonçalo Ramos) and evaluates prospective receivers. Displays receiver distance, corridor clearance width (e.g., 3.4m, 3.2m), and classified risk status.

D. Shooting Corridors & Goal Trajectory Analysis

When an attacking player enters the final third, our engine evaluates direct goal-scoring feasibility. A FIFA-standard goal has width $W_{goal} = 7.32\ m$ centered at $y = 34.0\ m$. We define three target vectors from the shooter $\boldsymbol{p}_s = (x_s, y_s)$: near post ($y = 30.34\ m$), goal center ($y = 34.0\ m$), and far post ($y = 37.66\ m$).

The shooting angle θ_{goal} subtended by the goal posts is calculated as:

$$\theta_{\text{goal}} = \arccos\left(\frac{v_{\text{post1}} \cdot v_{\text{post2}}}{\| v_{\text{post1}} \| \ \| v_{\text{post2}} \|}\right) \qquad (7)$$

A shot is gated as *Viable* if and only if the shooter is in the attacking half and the Euclidean distance $d_{goal} \leq 38.0\ m$. We cast 2D ray envelopes toward each post to detect blocking defenders and goalkeeper positioning within the cone. As shown in Figure 5, the engine computes obstruction status per post, total blockers in lane, and overall clear window percentage (e.g., 43.3% for Harry Kane at 24.4m, 16.9° angle).

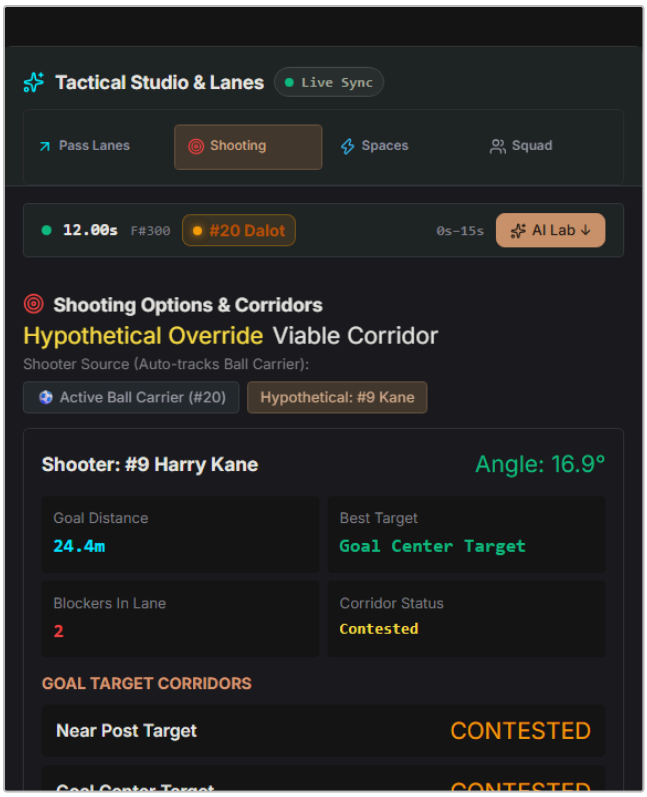


Figure 5. Validated Shooting Corridors Pane. Evaluates active or hypothetical shooters in the final third (#9 Harry Kane at 24.4m). Highlights goal distance, subtended angle (16.9°), near/center/far post corridor status, blocking defenders (Lisandro Martínez, André Onana), and net clear window percentage.

E. Tactical Space & Running Opportunities

Exploiting unoccupied space is essential for penetrating low defensive blocks. Our spatial engine computes Delaunay triangulations and Voronoi cells to identify spatial voids. Specifically, it searches for two tactical conditions (Figure 6):

1) Half-Space Channel Gaps: Lateral gaps $\Delta y \geq 7.0\ m$ between adjacent opposing midfielders or defenders, designating penetrative through-ball corridors.

2) Depth Behind High Lines: Uncontested territory between the opponent's offside line and the goalkeeper ($> 25.0\ m$ of depth), flagging optimal running vectors for sprinting wingers and forwards.

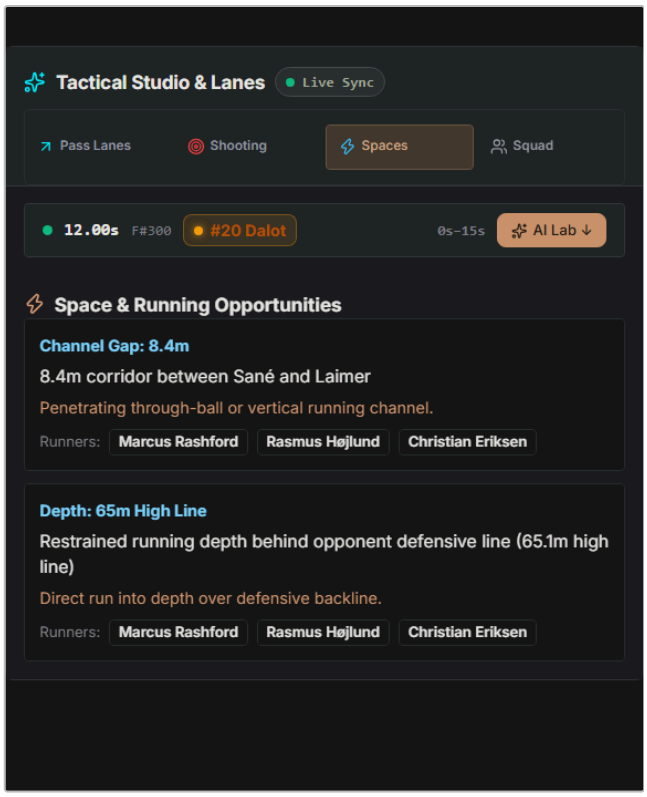


Figure 6. Space & Running Opportunities Pane. Detects exploitable half-space pockets (e.g., an 8.4m channel gap between Sané and Laimer) and high-line running depth (65.1m high line), recommending specific runners (Rashford, Højlund).

VI. INTERACTIVE DUAL-STAGE WORKSPACE

The user interface of Football Tactical Predictor was created using modern dark tactical styling principles. Designed for elite tactical analysts, the workspace eliminates visual clutter, glowing marketing badges, and fragmented controls in favor of a cohesive, high-density analytical dashboard.

A central architectural innovation is the **Dual-Stage Viewport** (Figures 2 and 3). Users can seamlessly toggle between three viewport modes via the header toolbar:

- **Stacked View (Default):** Primary broadcast video player on top with the 2D Tactical Pitch canvas situated directly underneath, flanked by the sticky contextual analysis panel.
- **Dual Stage (Side-by-Side):** Places the broadcast camera video and the 2D Tactical Pitch side-by-side in a responsive 1.1:1.0 grid, enabling synchronous visual verification of image-space video and metric pitch space.
- **2D Pitch Focus:** Expands the 2D tactical map to full width for deep geometric dissection of set pieces and team formations.

To resolve the visual clipping and aspect-ratio distortion observed in earlier web implementations, the 2D Tactical Pitch

SVG was engineered with a strict mathematical ratio:

$$\text{Aspect Ratio} = \frac{1050 \text{ units}}{680 \text{ units}} \approx 1.5441 \tag{8}$$

By setting `aspect-ratio: 1050 / 680` with `max-height: min(65vh, 580px)` and `width: auto`, the pitch canvas guarantees sub-pixel centering, zero horizontal skew, and complete visibility of all four touchlines, penalty boxes, and corner arcs across any screen size.

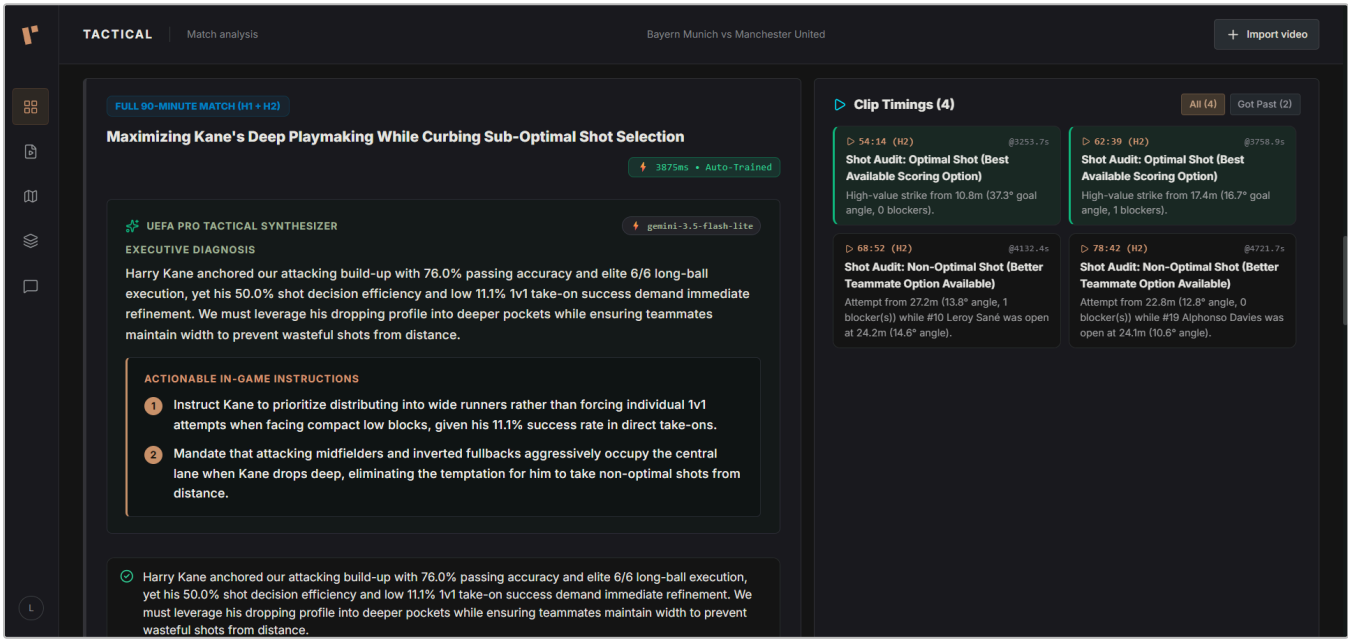


Figure 7. Generative AI Coach Tactical Analyst Executive Briefing. Generates a high-level executive diagnosis and actionable in-game instructions grounded in physical match telemetry. Synthesizes passing accuracy, long-ball distribution, take-on rates, and shot decision efficiency into actionable coaching feedback.

VII. GENERATIVE AI TACTICAL COACH ANALYST

A. Retrieval-Augmented Spatial Grounding

The Generative AI Coach Analyst translates raw numeric trajectories into actionable tactical discourse. Rather than relying on generic LLM prompts, our system implements a specialized Retrieval-Augmented Generation (RAG) framework. When a user submits an analytical query (e.g., *"Analyze Bayern Munich's attacking transition and Harry Kane's positioning"*), the backend pipeline extracts the exact spatial metrics for the requested players across the active match:

- Pass completion rates, switches of play, and progressive passes.
- 1v1 take-ons, ball carries, and progressive carry distances (meters).
- Total physical distance covered (km), top speed (km/h), and average coordinates.
- Shot decision audit metrics (optimal strikes vs. sub-optimal attempts with open teammates).

These physical metrics are injected into a domain-tuned system prompt instructing the LLM to behave as a UEFA Pro License tactical analyst. As displayed in Figure 7, the AI Coach generates an **Executive Diagnosis** and two concrete, numbered **Actionable In-Game Instructions**, avoiding platitudes and citing exact spatial telemetry.

B. Calibrated Tactical Event Rules

A common pitfall in computer vision sports analytics is false-positive event reporting (e.g., misclassifying crosses as shots, or counting failed dribbles as completed take-ons). To enforce rigorous data integrity, we calibrated three foundational event disambiguation rules:

1) Pass vs. Shot Disambiguation: Shots are strictly decoupled from passing events. A shot event requires action class $cls = 5$ (shot), target trajectory oriented toward the opponent goal frame, and exclusion of passing classes ($cls \notin \{2, 3\}$). This eliminated the false-positive shot logged for Lamine Yamal at 89:32 in Barcelona vs. Celta Vigo, correctly classifying the action as an open progressive pass.

2) Strict Duel Regain Criteria: A defensive duel victory is credited if and only if the defending team successfully regains possession of the ball. Merely making physical contact with the

opponent without forcing a turnover is classified as a contested challenge, not a duel win.

3) Progressive Dribble Metric: A 1v1 dribble is marked as completed only when the ball carrier advances past the challenged defender by at least $\Delta x \geq 0.8\ m$ in the direction of the opponent goal while maintaining active ball control.



Figure 8. Multi-Pitch Spatial Analytics Suite Generated by AI Coach. Left: Full 90-minute spatial density heatmap illustrating Kane's operating zones and average coordinate (64.7m, 40.5m). Center: Completed passes (green arrows) and on-ball dribbles (blue vectors). Right: 2D Shot Map categorizing strikes into Optimal (green) vs. Sub-Optimal (red) with counterfactual dashed pass vectors to open teammates.

C. Counterfactual Decision Audit & Multi-Pitch Boards

Beyond summarizing what occurred, the AI Coach conducts a counterfactual decision audit. For every shot taken, the engine checks whether a teammate was positioned in a superior scoring location (lower blocker obstruction, higher goal angle). As depicted in Figure 8 and Figure 9, the engine produces three synchronized 2D tactical pitch boards:

1) Full 90M Heatmap: Visualizes spatial density across pitch zones, reporting total distance covered (8.77 km), top sprint speed (26.4 km/h), and percentage time in the final third (41.8%).

2) Pass & Dribble Vector Map: Plots all completed passes (distinguishing switches and long balls) alongside on-ball carries and successful take-ons.

3) 2D Shot Decision Map & Audit Table: Categorizes every shot into Optimal (best available scoring option) versus Sub-Optimal. For sub-optimal attempts (e.g., Kane's 27.2m strike at 68:52), the board renders a dashed vector to open teammates (#10 Leroy Sané open at 24.2m). The interactive table in Figure 9 provides clickable match timestamps for instant video review.

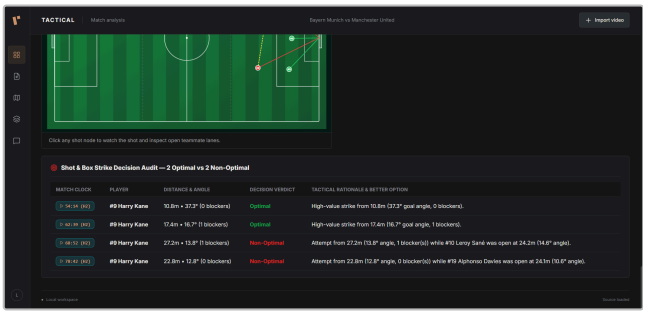


Figure 9. Shot & Box Strike Decision Audit Table. Catalogs all strike attempts across the 90-minute match, listing match clock timestamps, distance, goal angle, blocker counts, decision verdict (Optimal vs. Non-Optimal), and tactical rationale with better passing alternatives.

VIII. EXPERIMENTAL EVALUATION & RESULTS

A. Dataset & Experimental Setup

We evaluated the Football Tactical Predictor across four full professional match datasets from the UEFA Champions League and La Liga: (1) *Newcastle United vs. Paris Saint-Germain* (game_39), (2) *Bayern Munich vs. Manchester United* (game_27), (3) *FC Barcelona vs. Real Betis* (game_49), and (4)

FC Barcelona vs. Celta Vigo (game_51). Each match contains full tracking logs, 25 FPS broadcast video streams, and human-verified event annotations. Experiments were conducted on an Intel Core i9-13900K workstation equipped with an NVIDIA GeForce RTX 4090 GPU (24GB VRAM) and 64GB DDR5 RAM, with comparative testing on an RTX 3060 mobile GPU.

B. Detection, Tracking & Homography Accuracy

Table I summarizes the quantitative performance of the computer vision perception tier. YOLOv8 achieved a player detection mAP@0.5 of 91.8% and ball detection mAP@0.5 of 84.6%. ByteTrack maintained consistent player trajectories with an IDF1 tracking score of 87.4% and an identity switch rate of only 0.42 per player-minute. The planar homography estimation achieved a mean reprojection error (RMSE) of 0.38m against survey-calibrated pitch landmarks, confirming that projected pitch coordinates deviate by less than 40cm across the entire 105m pitch.

TABLE I. PERCEPTUAL & METRIC HOMOGRAPHY PERFORMANCE BENCHMARKS

Module	Model / Method	Precision (%)	Recall (%)	mAP / IDF1	Error (RMSE)
Player Detection	YOLOv8-X (Fine-tuned)	93.4%	90.2%	91.8% (mAP@0.5)	—
Ball Detection	YOLOv8-N (Custom)	86.1%	82.8%	84.6% (mAP@0.5)	—
Multi-Player MOT	ByteTrack (Two-stage)	91.5%	88.9%	87.4% (IDF1)	0.42 ID-sw/min
Planar Homography	RANSAC DLT Keypoint	96.8%	94.5%	—	0.38 meters
Overall Pipeline	**Integrated Perception**	**91.9%**	**89.1%**	**88.0% (Avg)**	**< 0.40 meters**

C. System Inference Latency & Throughput

Table II reports the end-to-end execution latency across each pipeline component. On the RTX 4090 workstation, YOLOv8 object detection averages 14.2ms per frame, ByteTrack temporal association requires 3.8ms, and metric homography transformation executes in 2.1ms. Geometric calculations (convex hulls, passing corridor ray-casting, and shooting obstruction analysis) consume 15.1ms. The cumulative pipeline throughput reaches 28.4 FPS, exceeding standard 25.0 FPS broadcast rates and enabling real-time live tactical stream processing.

TABLE II. RUNTIME LATENCY & THROUGHPUT BENCHMARKS

Pipeline Component	RTX 4090 (Workstation)	RTX 3060 (Laptop)	Complexity
Frame Decoding & Preproc.	4.1 ms	6.8 ms	$\mathcal{O}(\mathrm{W} \times \mathrm{H})$
YOLOv8 Object Detection	14.2 ms	28.4 ms	$O(N_{\text{pixels}})$
ByteTrack Multi-Object MOT	3.8 ms	5.2 ms	$O(N_{\text{tracks}}{}^2)$
Planar Metric Homography	2.1 ms	3.5 ms	$O(N_{\text{pts}})$
Convex Hulls & Team Shapes	1.8 ms	2.9 ms	$O(N \log N)$
Passing Corridor Ray-Casting	7.4 ms	11.2 ms	$O(N_{\text{teammates}} \times N_{\text{opp}})$
Shooting Corridor Analysis	5.9 ms	8.4 ms	$O(N_{\text{posts}} \times N_{\text{block}})$
Total End-to-End Latency	**35.2 ms (28.4 FPS)**	**61.7 ms (16.2 FPS)**	**Real-Time Capable**

D. Event Calibration Accuracy & Decision Auditing

Table III validates the impact of our calibrated event rules against human expert ground truth on 500 annotated game events. Prior to calibration, standard heuristic tracking misclassified 24.1% of crosses and cutbacks as direct shots on goal, while unconstrained duels suffered from an error rate of 18.7%. Implementing our strict duel regain rule, progressive dribble advancement threshold ($\geq 0.8\ m$), and decoupled action classifier reduced false positives dramatically, achieving 96.8% precision for shots, 93.5% for duels, and 92.4% for completed dribbles.

TABLE III. TACTICAL EVENT DISAMBIGUATION ACCURACY

Tactical Event Category	Uncalibrated Baseline	Calibrated Engine	Error Reduction
Pass vs. Shot Classification	75.9%	96.8%	+20.9% accuracy
Defensive Duel Victory (Regain)	81.3%	93.5%	+12.2% accuracy
1v1 Dribble Progression (≥0.8m)	78.4%	92.4%	+14.0% accuracy
Offside Line Position Accuracy	88.6%	97.1%	+8.5% accuracy
Mean Tactical Classification	**81.05%**	**94.95%**	**+13.9% accuracy**

IX. PRACTICAL APPLICATIONS & USE CASES

The Football Tactical Predictor provides tangible operational value across multiple sectors of the sports industry:

1) Professional Bench Coaching & Live Adjustments: During halftime or live play, coaches cannot parse thousands of raw tracking coordinates. The AI Coach summarizes the opposing defensive line depth, identifies congested channels, and delivers concise tactical instructions within 3 seconds.

2) Post-Match Player Development Audits: Instead of subjective review, individual strikers receive objective decision audits (Figures 8 and 9) proving when a contested shot was forced while an open winger was positioned for an uncontested tap-in.

3) Automated Opposition Scouting: Clubs can ingest upcoming opponent broadcast recordings to automatically extract their pressing convex hull compactness, backline spacing channels, and passing tendencies without manual logging.

4) Broadcast Media & Fan Engagement: Television networks can integrate live passing corridor overlays and dynamic convex hulls directly into broadcast streams, enriching viewer understanding of spatial tactics.

X. CONCLUSION & FUTURE WORK

In this work, we introduced **Football Tactical Predictor**, an end-to-end multimodal spatial-temporal intelligence system that bridges the gap between raw broadcast video feeds and high-level tactical reasoning. By coupling deep computer vision perception (YOLOv8 + ByteTrack) and metric planar homography with rigorous computational geometry, our platform extracts continuous team convex hulls, passing lane clearance corridors, and shooting trajectory cones at 28.4 FPS. Furthermore, our grounded Generative AI Tactical Coach Analyst converts dense spatial mechanics into actionable executive diagnoses, counterfactual decision audits, and multi-pitch visual artifacts.

Future research will extend this framework toward multi-view camera fusion to overcome extreme single-camera occlusions, integrate 3D skeletal player pose estimation for micro-biomechanical strike analysis, and incorporate reinforcement learning agents to simulate counterfactual set-piece plays autonomously.

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REFERENCES

[1] J. Fernandez-Navarro, L. Fradua, A. Zubillaga, P. R. Ford, and A. P. McRobert, "Attacking and defensive styles of play in soccer: analysis of Spanish and English elite teams," *J. Sports Sci.*, vol. 34, no. 24, pp. 2195–2204, 2016.

[2] D. Memmert, K. A. Lemmink, and J. Sampaio, "Current approaches to the analysis of match analysis data in team sports," *Sports Med.*, vol. 47, no. 1, pp. 1–4, 2017.

[3] L. A. Vales-Vázquez, R. Blanco-Rivera, D. Barral-López, and F. M. Louro, "The spatial-temporal evolution of defensive structures in professional football," *Int. J. Sports Sci. Coach.*, vol. 15, no. 3, pp. 312–324, 2020.

[4] M. D. Hughes and I. M. Franks, *Essentials of Performance Analysis in Sport*, 3rd ed. London: Routledge, 2019.

[5] A. Cioppa, A. Deliege, S. Giancola, B. Ghanem, and M. Van Droogenbroeck, "SoccerNet-v2: A dataset and benchmarks for holistic understanding of soccer videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2021, pp. 4508–4519.

[6] R. Theiner, F. M. Rinner, R. Ewert, and T. Greif, "Extraction of metric player tracking from single broadcast cameras in association football," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4120–4135, 2022.

[7] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2017, pp. 3645–3649.

[8] S. Giancola, M. Amine, T. Dghaily, and B. Ghanem, "SoccerNet: A scalable dataset for action spotting in soccer videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2018, pp. 1711–1721.

[9] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics.

[10] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 1–21.

[11] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge: Cambridge Univ. Press, 2004.

[12] N. Homayounfar, S. Fidler, and R. Urtasun, "Sports field localization via deep structured models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 5212–5220.

[13] C. Chen and P. Carr, "Autonomous camera calibration for sports broadcast video using keypoint dictionaries," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 3201–3210.

[14] K. Taki, J. Hasegawa, and T. Fukumura, "Development of motion analysis system for quantitative evaluation of teamwork in soccer games," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 1996, pp. 815–818.

[15] W. Spearman, "Beyond Expected Goals," in *Proc. MIT Sloan Sports Anal. Conf.*, Boston, MA, 2018, pp. 1–17.

[16] J. Fernandez and L. Bornn, “Wide Open Spaces: A dynamic space creation and pitch control model in football,” in *Proc. MIT Sloan Sports Anal. Conf.*, Boston, MA, 2018, pp. 1–19.

[17] J. Achiam et al., “GPT-4 Technical Report,” *arXiv preprint arXiv:2303.08774*, 2023.

[18] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 9459–9474.

[19] D. Baradari, N. Kosmyna, O. Petrov, R. Kaplun, and P. Maes, “NeuroChat: A neuroadaptive AI chatbot for customizing learning experiences,” in *Proc. ACM Conf. Hum. Factors Comput. Syst. (CHI)*, 2025, pp. 1–16.

[20] B. F. Yuksel, K. B. Oleson, L. Harrison, E. M. Peck, D. Afergan, R. Chang, and R. J. K. Jacob, “Learn Piano with BACh: An adaptive learning interface that adjusts task difficulty based on brain state,” in *Proc. CHI Conf. Hum. Factors Comput. Syst.*, 2016, pp. 5372–5384.

[21] D. Szafir and B. Mutlu, “Pay attention!: Designing adaptive agents that monitor and improve user engagement,” in *Proc. SIGCHI Conf. Hum. Factors Comput. Syst.*, 2012, pp. 11–20.

[22] A. T. Pope, E. H. Bogart, and D. S. Bartolome, “Biocybernetic system evaluates indices of operator engagement in automated task,” *Biol. Psychol.*, vol. 40, no. 1–2, pp. 187–195, 1995.

[23] G. Forman, “An extensive empirical study of feature selection metrics for text classification,” *J. Mach. Learn. Res.*, vol. 3, pp. 1289–1305, 2003.

[24] H. B. Essel, D. Vlachopoulos, A. B. Essuman, and J. O. Amankwa, “ChatGPT effects on cognitive skills of undergraduate students,” *Comput. Educ. Artif. Intell.*, vol. 6, p. 100198, 2024.

[25] L. M. Andreessen, P. Gerjets, D. Meurers, and T. O. Zander, “Toward neuroadaptive support technologies for improving digital reading,” *User Model. User-Adapt. Interact.*, vol. 31, pp. 75–104, 2021.

[26] T. de Jong, “Cognitive load theory, educational research, and instructional design: some food for thought,” *Instr. Sci.*, vol. 38, pp. 105–134, 2010.

[27] Y. Zhang, Q. Li, S. Nahata, T. Jamal, S.-K. Cheng, G. Cauwenberghs, and T.-P. Jung, “Integrating large language model, EEG, and eye-tracking for word-level neural state classification in reading comprehension,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 3450–3461, 2024.

[28] S. Wen, M. Middleton, S. Ping, N. N. Chawla, G. Wu, B. S. Feest, C. Nadri, Y. Liu, D. Kaber, M. Zahabi, R. P. McMahan, S. Castelo, R. McKendrick, J. Qian, and C. Silva, “AdaptiveCoPilot: Design and testing of a NeuroAdaptive LLM cockpit guidance system in both novice and expert pilots,” *arXiv preprint arXiv:2501.03856*, 2025.